

9

THE EMERGENCE OF DESAKOTA REGIONS IN ASIA: EXPANDING A HYPOTHESIS

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The symbiosis of urban and rural in Megalopolis, creating new and interesting patterns of multiple-purpose land use over large areas, gives to the region a rather unique character. Like the downtown business districts with powerful skylines, this aspect of Megalopolis will probably be repeated in slightly different but not too dissimilar versions in many other regions of the rapidly-urbanizing world.

—Jean Gottmann¹

Paddy has developed a strikingly similar landscape, broadly similar from the Ganga to the Yangtze ... but no other way of life ... has led to the evolution of a cultural system so stable and permanent as that associated with the great paddy-plains of Monsoon Asia.

—O. H. K. Spate and A. I. A. Learmonth²

This chapter is a preliminary exploration of the emergence or what appear to be new regions of extended urban activity surrounding the core cities of many Asian countries.³ The ideas to be explored must be placed in the framework of the overall patterns of urbanization at a global and regionalized level, which are predicting a continuing increase in the proportion of the world's urban population. By the year 2020, the UN Centre for Human Settlements predicts that more than 57 percent of the world's population will be living in urban places.⁴ This population will be unevenly urbanized, with levels of

urbanization at almost 77 percent in developed countries and 53 percent in developing countries. Within the developing countries, the contrast will be even greater, with Latin America 83 percent urbanized, and Africa and Asia close to 50 percent. However, Asia's urban population will account for a very large portion of that of the developing countries. Bangladesh, India, China, Indonesia, and Pakistan together will contain 34 percent of the developing countries' urban population.

These United Nations predictions are largely based on assumptions concerning the growth of population in places defined as urban. The predictions are calculated using growth rates reflecting performance in previous decades. When projected forward, they appear to suggest a successful shift to urbanized societies and a repetition of patterns of the more developed countries. As Ginsburg has commented about urbanization in the United States, "This condition reflects the progression of the ... space-economy to a state of what one might consider 'maturity,' that is, to a condition whereby areas possessed of substantial comparative advantage ... would be drawn effectively, through improved transportation networks, into the national geographic structure."⁵ The implication for the urban systems of the largest developing countries is that a continued growth will create immense cities of about 16 million to 30 million. However, this may not be the only possible outcome for Asian urbanization.

The purpose of this chapter is to draw upon earlier ideas put forward by McGee and Ginsburg to challenge this particular view of the urban transition.⁶ The Ginsburg-McGee position essentially argued that in the Asian context, the conventional view of the urban transition—which assumes that the widely accepted distinction between rural and urban will persist as the urbanization process advances—needs to be reevaluated. Distinctive areas of agricultural and non-agricultural activity are emerging adjacent to and between urban cores, which are a direct response to preexisting conditions, time-space collapse, economic change, technological developments, and labor force change occurring in a different manner and mix from the operation of these factors in the Western industrialized countries in the nineteenth and early twentieth centuries.

To elaborate further, the conventional view of the urban transition is inadequate in three respects. First, it is too narrow in its view that the widely accepted spatial separation of rural and urban activities will persist as urbanization continues.

Second, it is inadequate in its assumption that the urbanization transition will be inevitable because of the operation of "agglomeration economies" and comparative advantage, which are said to facilitate the concentration of the population in linked urban places. The emergence of such a system was described by Jean Gottmann in 1961 as a "megalopolis" in which, when applied to the northeastern United States, the population was largely concentrated in the urban and suburban areas, but interspersed with areas of low population density used for agriculture and as leisure spaces by the population of the

megapolitan areas.⁷ In many parts of Asia, the spatial juxtaposition of many of the larger city cores within heavily populated regions of intensive, mostly wet-rice agriculture based on a mixture of “skill-oriented” and “mechanical” technological inputs has created densities of population that are frequently much higher than in the suburban areas of the West.⁸ This juxtaposition permits demographic densities similar to urban areas over extended zones of intensely cultivated rural areas located adjacent to urban cores. The considerable advances in transportation technology, particularly in relatively cheap intermediate transportation technology, such as two-stroke motorbikes, greatly facilitate the circulation of commodities, people, and capital in such regions, creating, in turn, large mega-urban regions.

Third, the Western paradigm of the urban transition, which draws its rationale from the historical experience of urbanization as it has occurred in Western Europe and North America in the nineteenth and twentieth centuries, is clearly not neatly transferable to the developing countries’ urbanization process. The uneven incorporation of these Asian countries into a world economic system from the fifteenth century onward created divergent patterns of urbanization, which reflect the different interactions between Asian countries and the world system.⁹ For example, the British, French and Dutch all developed the productivity of wet-rice agriculture in Southeast Asia.¹⁰ In a similar manner, Japanese rule in Korea and Taiwan further accentuated the monocrop rice characteristics of parts of these countries as sources of supply for Japan’s prewar empire. Geopolitical events determined that both these countries emerged into “fragile” independence with high rural densities and low levels of urbanization. On the contrary, British intervention in Malaysia created an urban system oriented to the production of export products on the west coast away from the heavily populated rice bowls of Kedah and Kelantan, limiting the possibilities of an emergent mega-urban region.

Because of these inadequacies in the conventional view, the concept of the “urban transition” needs to be positioned within a broader paradigm of the transition in the space-economy of countries. Such a paradigm would include (1) a heightened sensitivity to the historical elements of the urban and agrarian transition within specific countries; (2) an appreciation of the ecological, demographic and economic foundations of the urban and agrarian transition; (3) an investigation of the institutional components, particularly the role of the state in the development process; (4) a careful evaluation of the transactional components within given countries including transport, commodity, and population flows; and (5) a broad understanding of the structural shifts in the labor force reflecting economic change. Essentially, such an approach is an attempt to investigate the manner in which particular sets of conditions in one place interact with broader processual change. It is not so much concerned with the contrast between rural and urban as the space-economy changes but focuses instead on the interactions within the space economy as they affect the emergence of particular regions of economic activity. This view has important implications for policy making in the Asian context, for it poses a challenge to sectoral approaches to development planning.

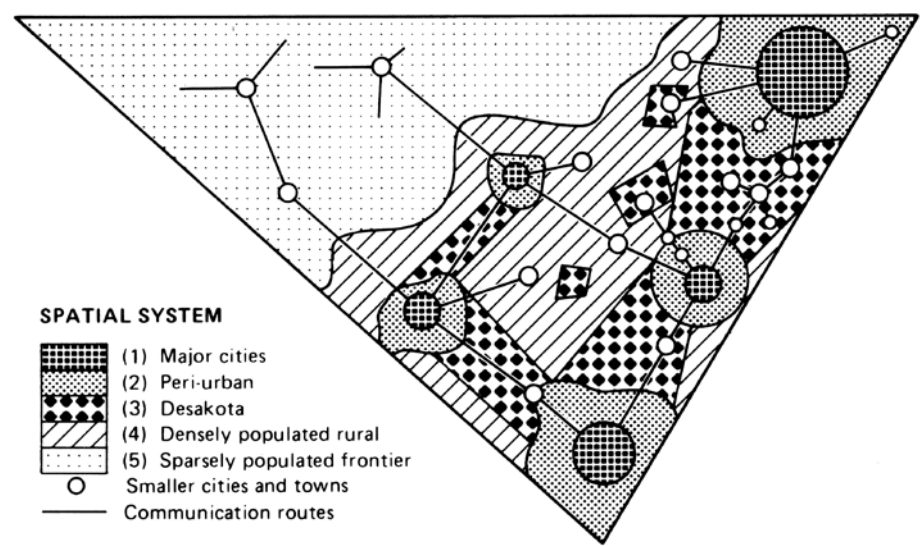
Definitions and Parameters

Since this assertion is quite challenging to those who have vested interests in the persistence of the urban-rural paradigm, it is necessary to spell out in some detail the definitional components of this broader view of the “space-economy transition.”

Figure 9.1 presents a model of the spatial configuration of a hypothetical Asian country, which I will label Asiatica Euphoria for the purposes of this exercise.¹¹

In this example, five main regions of the spatial economy are identified as follows:

1. The major cities of the urban hierarchy, which are often dominated in the Asian context by one or two extremely large cities.
2. The peri-urban regions, which are those areas surrounding the cities within a daily commuting reach of the city core. In some parts of Asia, these regions can stretch for up to thirty kilometers away from the city core.
3. The regions labeled *desakota*, which are regions of an intense mixture of agricultural and non-agricultural activities that often stretch along corridors between large city cores.¹² These regions were previously characterized by dense populations engaged in agriculture, generally but not exclusively dominated by wet-rice.
4. Densely populated rural regions, which occur in many Asian countries, particularly those practicing wet-rice agriculture.



9.1 Spatial configuration of a hypothetical Asian country

5. Finally, the sparsely populated frontier regions found in many Asian countries that offer opportunities for land colonization schemes and various forms of agricultural development.

The model of the spatial economy is, of course, static and must change as the economy changes. The pace and characteristics of this settlement transition vary from country to country, reflecting the features of socioeconomic change at the macro level. The role that the growth of metropolitan cores and the *desakota* process play in this transformation is of major importance. The mega-urban regions that emerge often incorporate two large urban cores linked by effective transportation routes. These regions include the major cities, peri-urban zones, and an extensive zone of mixed rural-urban land use along such routes. Travel time between any two points in a region would probably be no more than three to four hours, but in most cases is considerably less. Mixed economic activities may also occur in villages in these zones, which are less accessible and where economic linkages are more reliant on social networks.

It should be stressed that this model of the transition of the space-economy is not intended to be universally applicable, but to fit the situation where one or more urban cores are located in densely settled peasant rural areas.¹³ There may also be cases where the urban cores are located in lightly populated regions of plantation agriculture as in the case of Kuala Lumpur in Malaysia. The contrast between the two agro-economic systems as they are reflected in socioeconomic systems, export trade, and class relations is not a new theme. It has been used by Baldwin for a theoretical exposition of patterns of development in newly settled regions, by Dowd to explain the differences in the settlement patterns of the American west and south, and by Morse to explain the different urban systems that evolved in the “hacienda” and “plantation” regions of Latin America.¹⁴ These writers are not suggesting that a particular urban system results from a preexisting agro-economic system, but rather that the existence of these agro-economic systems provides the possibility for the emergence of certain urban systems and regions.

In the Asian context, the existence of high-density agricultural regions adjacent to large urban cores offers an opportunity for a particular form of mega-urban region to emerge. Their existence does not ensure the inevitability of the emergence of such regions. These will result from, for example, the policies of private and public sectors, the form of economic growth, and the position of the urban core relative to international connections. In the case of the Republic of Korea, with a precondition of high-density rice regions, the government adopted a strategy of concentrating on industrialization rather than agriculture, which led to slow growth in rural income and a release of surplus rural population into urban-based industrialization. Thus, South Korea was characterized by a metropolis-dominated urban hierarchy. By contrast, in a region of similar pre-existing rural densities such as Jogjakarta in Java, in a slow-growth situation there are only limited possibilities for drawing off surplus rural population to urban centers in other parts of the country, and the

rural inhabitants engage in an intense mixture of non-agricultural and agricultural activity that permits survival but does not increase income.

There are at least three types of spatial economy transition occurring in Asia in regions that have the prerequisite of the historical evolution of high-density, mostly rice-growing agroeconomic niches.

First, there are those countries that have seen a decline in rural settlement, land use and agricultural population as the population has moved to urban centers. In such countries, agricultural land use may remain important as a reflection of government land or agricultural protection policies. This pattern has been associated with overall increases in income and productivity in which rural populations fall well behind those of urban areas. South Korea and Japan are examples of such a spatial economy transition. These countries will be labeled *desakota* Type I or *konjuka*, the Japanese term for landscapes that have a mixture of small farm plots, residences and industry. Such regions are characterized by rural landscapes in which most of the economically active work is in non-agricultural activities.¹⁵

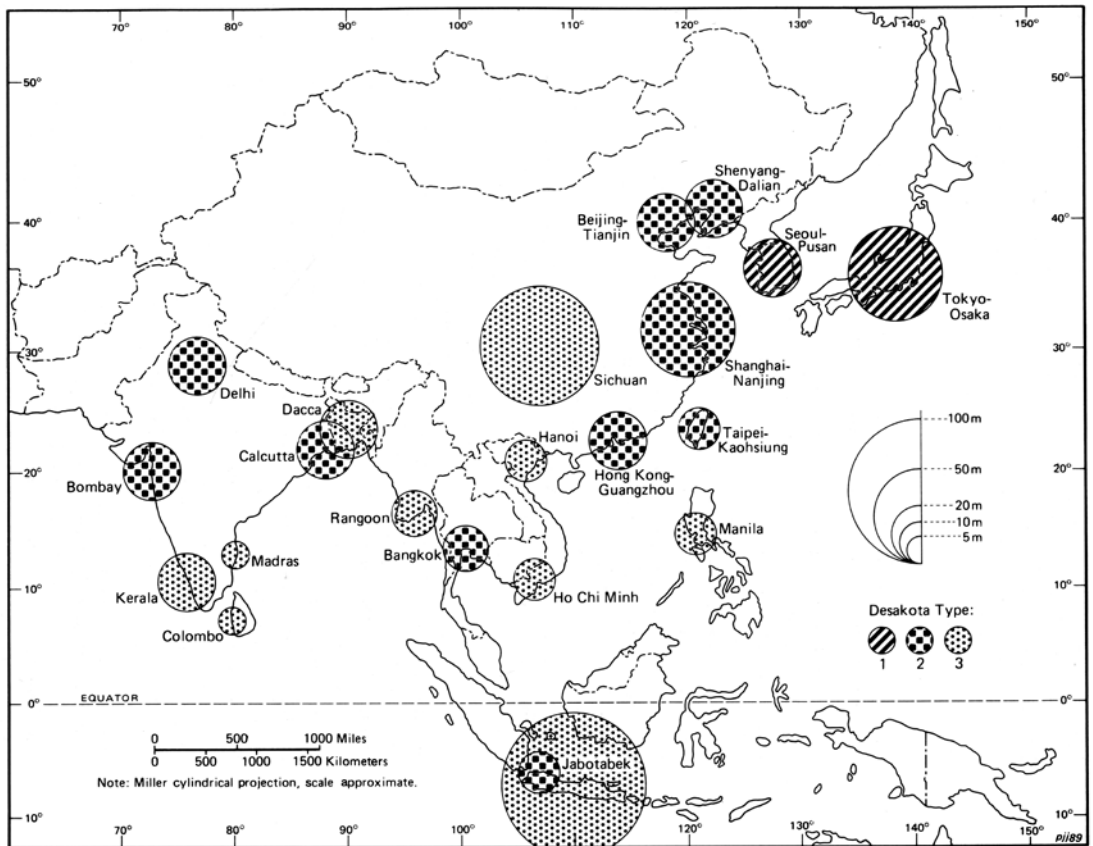
Second, there are those regions in which, over varying periods of time, productivity gains in agriculture and industry, and secular shifts from agricultural to non-agricultural activities are focused particularly on the urban cores and adjacent regions. These changes are linked to rising household income, improved transportation linkages, and improved infrastructures. Examples occur in regions such as Nanjing-Shanghai-Hangzhou, the Central Plains of Thailand, the Taipei-Kaohsiung corridor, the Calcutta region and Jabotabek in Java. These may be identified as *desakota* Type 2, and are characterized by rapid economic growth compared to other regions of the country.

Third, there are those regions of high density in which economic growth is slow. Often such regions are located close to secondary urban centers that have slow economic growth, and are characterized by continuing high population growth, surplus labor and persistent low productivity in both agriculture and non-agriculture. Examples are the Jogjakarta region in Java, Kerala in South India, Bangladesh and the Sichuan Basin in the interior of China. These regions will be labeled *desakota* Type 3, and are characterized by slow growth of income and involuntary economic activity.¹⁶

The Emergence of the Extended Metropolitan Region in Asia

Figure 9.2 shows the location of the core regions in Asia as grouped into the three main *desakota* types. First are those countries that have experienced a rapid transformation of the spatial economy in terms of rural-to-urban shift in population, although agricultural land use may remain quite persistent. Japan and South Korea are the most prominent examples.¹⁷

Second, there are those regions of countries that have experienced a rapid change in their economic features in the past 30 years. An example is the Taipei-Kaohsiung corridor of Taiwan, which has experienced a declining proportion of agricultural employment from 56 to 20 percent between 1956 and 1980, and a concurrent growth of industrialization. Speare and his colleagues have estimated that the growth of small to medium-sized industries in rural areas slowed the growth rate of cities by 6 percent in the 1960s and 1970s.¹⁸ At the same time, this region was characterized by a decline of staple crops as a proportion of the total agricultural value of production. Thus, over the past 30 years, although the production of rice has increased considerably, the share of rice as a proportion of gross agricultural receipts has dropped from 50 percent in 1950 to 34 percent in 1980. At the same time, other agricultural products have increased from 20 to 36 percent, and vegetables and fruit from 7 to 20 percent. More recently, there has been a rapid increase in fish farming (prawns), chicken rearing and other forms of capital-intensive agribusiness. This shift has led to a significant change in the pattern of female employment, with a decline from 52.5 percent (1965) in primary industry to 16 percent in 1980, and an increase in secondary industry from 18.2 to 43.7 percent.¹⁹ Similar patterns are being exhibited in the Bangkok-Central Plains region of Thailand and the four major coastal zones of China. These regions,



9.2 Growth of core areas in Asia

with only 12.5 percent of China's population, accounted for 46.4 percent of the value of industrial production and 13 percent of the value of agricultural output in 1986.

Third, there is a type of region that bears some spatial and economic resemblance to Type 2 but is characterized by changes that occur because of high population growth and slower economic growth. This situation results in the persistence of underemployment and self-employment in unpaid family work and enterprises. In such regions there may be a juxtaposition of elements of Type 1 and Type 2, producing a highly dualistic economic structure. Thus, technological inputs in agriculture may cause labor shedding and an increase in non-agricultural activities in the rural areas adjacent to urban cores. This phenomenon has been recorded in areas such as Kerala and Tamil Nadu. Although there is often some growth of small industry and other income opportunities, these regions are characterized by the persistence of low incomes, which reflect the slow structural transition in the allocation of labor. In some cases, regions continue to absorb population into agricultural areas (for instance, the Sichuan Basin) where non-agricultural employment has remained fairly static over the past forty years while the population has almost doubled in size.²⁰

Conditions and Processes Underlying the Emergence of New Zones of Economic Interaction: *Desakota*

Given the diversity of these transitions, it may well be argued that the *desakota* have little in common with one another. However, certain common conditions and processes occur in these regions.

First, virtually all these regions are characterized by densely populated, small-holder cultivation agriculture, commonly wet-rice, which involves careful water management and agronomic practices.²¹ Densities frequently approach 1,000 people per square kilometer of cultivated land. Historically, Oshima has argued, the pronounced seasonality and the intense labor of planting and harvesting have had two results: increasing population density, which has led to increasingly smaller farm plots, and uneven seasonal demands for labor inputs. During the off-season, "This dense population must look for off-farm employment since most farms are too small to generate enough income to live on ... Monsoon rice-farming never became separated from nonagriculture as it was in the West where cropping came to be combined with animal husbandry in large capitalistic farms."²² He goes on to argue that in the prewar centuries "monsoon agriculture kept large masses of workers tied down in rural areas, rendering the large labor supply inflexible."²³ In a pre-Green Revolution situation, the capacity of the population of rice-growing areas to increase and still produce enough rice to avoid famine even with very limited infrastructural investment is the basic explanation that Geertz provides for the growth of Java's population from 5 million to 25 million in 10 decades of slow economic growth.²⁴ This argument can be applied to many of the other Asian rice bowls in India, China, Japan, Korea and Southeast Asia. Bray reinforces this point with the following comment:

The organization of resources typical of a “skill-orientated” technology such as intensive rice farming dovetails very neatly with petty commodity production, which requires very little capital to set up a family enterprise and absorbs surplus labour without depriving the farm of workers at times of peak demand. It can be expanded, diversified or contracted according to market demands, but the combination with the rice farm guarantees the family subsistence. The products can be conveniently conveyed to national markets by merchants, who pay the villagers for their labour and often provide raw materials as well as information on the state of the market.²⁵

Thus, it is possible to argue that the labor force of these rice-bowl areas was “culturally” prepared to commit its labor to various forms of “new” non-agricultural activity.

Second, in virtually all the rice-bowl areas there were large cities or clusters of cities such as Calcutta, Shanghai, Bangkok and Guangzhou that provided both opportunities for seasonal labor and important markets for rural rice and other products. The linkages with these cities were important for the surrounding rural areas for cultural and economic reasons.

Third, these regions were frequently characterized by a well-developed infrastructure of roads and canals that allowed an intense movement of commodities and people. Indeed, one is constantly reminded of the importance of water systems for these regions as the fundamental sustainers of the ecological system. Colonial impact, whether by Britain, Japan or Holland, did little to change these patterns. Indeed, by the provision of political stability and investment in infrastructure the colonial powers frequently enhanced the preconditions for growing populations.

Fourth, by the early 1950s, all of these regions were large, cheap labor reservoirs waiting to be tapped by state, international and private capital investment. The manner in which this labor was incorporated into non-agricultural activity varied markedly from country to country and region to region. Most successful were Japan, Taiwan and South Korea, where both industrialization and agricultural growth led the way in Asia. In these three countries, efforts were made to increase rural income through higher yield, guaranteed prices, diversification into non-rice crops, and increased opportunities for employment in rural industries. This effort was aided by physical infrastructure improvements such as rural roads, electrification, consolidation and irrigation. Increased mechanization released labor, particularly women, to work in industry. Institutional changes, particularly land reform and the introduction of higher yielding rice varieties, were crucial in this process. In addition, the state invested in major transportation linkages such as freeways and electrified railways, which pulled these regions closer to the urban cores. Of course, these processes did not prevent the movement of labor out of agriculture into non-agriculture and urban centers, which accelerated in Japan in the 1960s, in Korea in the 1970s, and in Taiwan in the 1980s. But political and institutional imperatives, particularly the need

for food self-sufficiency, encouraged the persistence of rice farming and created *konjuka* landscapes.

In other countries, these processes moved more slowly, partly because the institutional changes of the “Three Tigers” could not be implemented. The Philippines is an excellent example of initial success and then slowdown because of bad government. In the late 1970s, however, other regions in Asia began to exhibit enough of the features responsible for the successes in Japan, Korea and Taiwan to achieve considerable growth. For example, the four regions of Hong Kong, Guangzhou-Macao, Nanjing-Shanghai-Hangzhou, Beijing-Tianjin-Tangshan and Shenyang-Dalian have exhibited all these features since 1978. Wang and Veeck have shown how remarkably these regions fit the model of *desakota*.²⁶ Other studies find the same processes occurring in the Central Plains of Thailand and even in the extended Calcutta Metropolitan Region, which has long had the reputation of being one of the main centers of world poverty.²⁷ Rice-bowl areas such as Kerala, Bangladesh, Jogjakarta and the Sichuan Basin, less favorably positioned from the viewpoint of state and private investment, exhibit a persistence of low income and continuing pressure of population on available resources.

Fifth, all the *desakota* regions were characterized by highly integrated “transactive” environments in terms of movement of people and commodities, for example. In many cases, the dense network of rural canals and waterways was central to this integration prior to the Second World War. The onset of technological developments in intermediate transportation since the 1950s has greatly accelerated this process.

Finally, it is important to acknowledge the role of the expansion of the global economy and the international division of labor, which create a situation where national governments responsible for gigantic, cheap labor pools have adopted different policies with respect to permitting or encouraging their countries’ labor to be tapped for national and international industrial growth. In this respect, the location of *desakota* Type 1 and 2 regions adjacent to large cities and transportation points has been particularly important. From the viewpoint of many investors, investment in industry in these regions is cheaper for virtually all the factors of production, and they are able to avoid some of the diseconomies that exist in the large urban zones. Thus, these regions are important areas of subcontracting for a portion of the industrial production process.

To summarize, the regions designated as *desakota* have six main features.

1. They have been or are characterized by a large population engaged in small-holder cultivation of rice, which before the Second World War conducted considerable interaction through accessible transportation routes.

2. They are generally characterized by an increase in non-agricultural activities in areas that have previously been largely agricultural. These non-agricultural activities are very diverse, and include trading, transportation and industry. The increase in non-agricultural activity is characterized by a mixture of activities, often by members of the same household. Thus, one person may commute to the city to work as a clerk, another engage in farming, a third work in industry, and another find employment in retailing in the *desakota* zone. This mixture creates a situation in which the economic linkages within such a region may be as important as the dominance of the large cities in the megalopolis that draw the surrounding regions into their orbit.
3. The *desakota* zones are generally characterized by extreme fluidity and mobility of the population. The availability of relatively cheap transport such as two-stroke motorbikes, buses and trucks has facilitated relatively quick movement over longer distances than could be covered previously. Thus, these zones are characterized not only by commuting to the larger urban centers but also by intense movement of people and goods within the zone.
4. The *desakota* zones are characterized by an intense mixture of land use with agriculture, cottage industry, industrial estates, suburban developments and other uses existing side by side. Such a mix has both negative and positive effects. Agricultural products, particularly industrial crops, have a ready market, but the waste of industrial activity can pollute and destroy agricultural land. On the whole, these zones are much more intensely utilized than the American megalopolis, with regard to which Gottmann commented on the amount of woodland and recreational areas that exist. In the *desakota* zones of Asian countries, population pressures place greater demands on the available space.
5. Another feature of the *desakota* zones is the increased participation of women in nonagricultural labor. In part, this feature is associated with a demand for female labor in industry, domestic service and other activities, but it is also closely related to changing patterns of agricultural production in the *desakota* regions. Generally, agricultural production shows a shift from monocrop grain cultivation to increased diversity with production of livestock, vegetables and fruit, sometimes for national and interregional consumption.
6. Finally, *desakota* zones are to some extent “invisible” or “gray” zones from the viewpoint of the state authorities. Urban regulations may not apply in these “rural areas,” and it is difficult for the state to enforce them despite the rapidly changing economic structure of the regions. This feature is particularly encouraging to the “informal sector” and small-scale operators who find it difficult to conform to labor or industrial legislation.

In essence, then, the central processors that shape these regions are the dynamic linkages between agriculture and non-agriculture, and investment seeking to utilize cheap labor and land within a distinctive agroecological setting. Ranis and Stewart have identified how expansion in agricultural output leads to an expansion in other activities and, conversely, how additional non-agricultural activity in the rural areas provides opportunities and incentives for raising agricultural productivity.²⁸

In the *desakota* regions that show the greatest increase of both agricultural and nonagricultural income, there is a general rise in household income. Depending on the expenditure decisions of households (savings/expenditure ratios), there may be an increase in demand for local supplies, goods from urban centers and imports. Ranis and Stewart show significant variations in different regions. Thus, in Taiwan an increase in agricultural income in rural areas was associated with increased nonagricultural production-related employment and high increases in all linkages. In the Philippines, slower rates of agricultural production associated with a large population increase led to an increase in low-income non-agricultural employment and limited linkages with larger markets. The Philippine situation is typical of the “involuntary” and “distress” features of surplus labor markets first discussed by W. Arthur Lewis.²⁹

Questions Concerning the *Desakota* Regions in Asia

The regions where the *desakota* processes are in full operation are clearly an important part of the “settlement transition” in Asia. For example, Zhou reported that the four key economic regions of China (Nanjing-Shanghai-Hangzhou, Hong Kong-Guangzhou-Macao, Beijing-Tianjin-Tangshan and Shenyang-Dalian) with approximately 12 percent of the nation’s population in 1986-87 were responsible for 47 percent of its industrial output. These four key economic regions play a crucial role in the current phase of rapid economic growth in China. Similar arguments were presented by Liu and Tsai for Taiwan. Thus, there is a great deal of support for the assertion that these regions are highly significant foci for the development process. However, further data are needed to support the assertion, particularly with respect to the following questions:

In what manner are these new economic regions different from the “zones of urban influence” that are well established in the urban transition literature? In other words, if these zones are simply a greater areal extension of the “peri-urban” region of large cities that have been brought about by space-time collapse and transportation improvements, then what is different about them from the so-called peri-urban regions? In fact, the relationship between the urban cores and the adjacent regions is important to the sustainability of these regions, and therefore any study of these regions must investigate the urban core functions and the relationships with the surrounding regions.

Can economic data be collected to show processes operating that facilitate certain types of economic activities in these regions? For instance, are there diseconomies associated with large cities that facilitate the location of economic activities in such zones? Is the legislative environment in such zones more permissive (are taxes lower?) for certain types of activities? Are wage rates persistently lower? Are the social overhead costs lower in such regions? In other words, are there economic benefits that facilitate the growth of economic activities in these regions that may be a “growth generator” in the development process?

Are such regions likely to exhibit short-term persistence (up to 50 years), or are they simply a transitory phenomenon? Some evidence presented by Liu and Tsai suggests that in Taiwan the persistence of such a region, at least in the corridor between Taipei and Kaohsiung, is questionable. They show evidence that the population is concentrating in the northwest quadrant of Taiwan as the industrial and services characteristics of the economy change. This shift is associated with a decline in agricultural employment. Is there some point in the development process at which the agglomerative tendencies will take over? This type of question has important policy implications for countries such as China or India as their economies develop.

Issues of Policy Formation

Prevailing policy prescription for macrointervention in the spatial shift of population during development is broadly polarized between acceptance of big-city growth representing the “rational” development of economies of scale and agglomeration, and arguments in favor of the development of “small” and “intermediate” towns in the urban hierarchy, decentralization of industry, and frontier development.³⁰ Current developments raise the question of whether *desakota* regions represent a viable “middle” policy option and, if so, what kinds of policies need to be adopted. A range of “problems” result from the growth of urban cores and the *desakota* regions adjacent to them, including environmental degradation, waste removal and adequate delivery of social welfare. Before the issue of some form of “middle planning option” is tackled, it will be necessary for the governments of the region to deal with the fundamental issue of the usefulness of a rural-urban distinction. In reviewing the debate on “urban bias,” Harriss and Moore explain this point as follows:

There are two themes central and common to their [“urban bias” analysts] work. First, in attempting to explain national level patterns of economic resource allocation within a political economy framework, they use the concept of economic sectors, mainly the rural/agricultural versus urban/industrial categorization. Second, they suggest in varying degrees that the way in which sectoral conflicts influence the allocation of economic resources through state action has been the prime cause of slow rates of economic growth (and in Lipton’s work, of growth biased against the poor) in developing countries since the Second World War.³¹

But what happens if the rural-urban dichotomy ceases to exist, as in the case of *desakota* zones? Then the whole policy debate on urban and rural allocation of resources becomes fuzzy and meaningless, unless one accepts Lipton's argument that "rural-urban is not a categorization of space alone. To see it like that is to underpin an incorrect, absolute distinction between geographical (residence, density), occupational, sectoral and class categorizations of households."³²

All societies have working spatial definitions of urban and rural areas. These definitions are highly variable from country to country and are often changed. However, most definitions have some common elements such as size and political definition. For instance, in some countries all gazetted towns exceeding a population of ten thousand are regarded as urban; everything else is rural. In fact, it is more important to know two important economic pieces of information:

1. What is the contribution of agricultural and non-agricultural activities to the GDP of a given spatial unit (nation, province, and so on)?
2. What is the proportion of the working labor force employed in agricultural and non-agricultural work in a given spatial unit?

If this information were available over given time periods, it would be possible to develop a more precise definition of urban and rural areas.

One could conceive a rather simple matrix constructed at the level of small administrative units that would allow a fourfold spatial division of a country on a continuum from the most urban spatial unit to the most non-urban spatial unit (Figure 9.3).

Assuming some ideal statistical base, this type of analysis would enable the estimation of the contribution of the urban spatial units to the GDP, as compared to the non-urban spatial units. Temporal data, if available, would permit the assessment of the relative contribution of urban and non-urban areas to the GDP through time as well as the differences in the labor force over time. This kind of information would provide vital feedback to the government in assessing the spatial impact of its investment policies. Unfortunately, few developing societies possess data that can be analyzed in this way, relying instead on macrodata that conceal these significant differences between urban and non-urban areas.

There appear to be six priorities for many Asian countries if they are to develop pragmatic strategies that attempt to recognize the importance of the *desakota* regions. First, the government will have to make some significant decisions with respect to agricultural policy. All indications are that in most Asian countries, the "agricultural issue" is of central importance to *desakota* regions. The problem revolves around the need for a sufficient supply of foodstuffs and the "cultural" demand for agricultural activity in Asian countries.

1 Spatial units with <i>more than 50 percent</i> • contribution by non-agricultural economic activities to spatial unit GDP • of the labor force in non-agricultural activities	3 Spatial units with <i>less than 50 percent</i> • contribution by non-agricultural economic activities to spatial unit GDP and <i>more than 50 percent</i> • of the labor force in non-agricultural activities	most urban least urban
2 Spatial units with <i>more than 50 percent</i> • contribution by non-agricultural economic activities to spatial unit GDP and <i>less than 50 percent</i> • of the labor force in non-agricultural activities	4 Spatial units with <i>less than 50 percent</i> • contribution by non-agricultural economic activities to spatial unit GDP; • of the labor force in non-agricultural activities.	

9.3 Spatial units and economic outputs

The crucial issue is how long rice growing, for example, will persist in *desakota* regions as economic growth proceeds. In the cases of Japan, Korea and Taiwan, it appears that policies will eventually reduce the role of rice in *desakota* areas. In other cases, labor is being released from rice growing as a result of technological changes over which governments apparently have little control.

Second, Asian governments will seriously need to consider in what manner the release of labor from agricultural labor pools is going to occur. Policies that slow down geographical relocation and foster *in situ* development should be given priority. Given the already high levels of development in these regions, they should not be expensive. Of course, the problems of the more “traditional” regions of *desakota*, such as Sichuan, are more intractable and should be tackled by more conventional development inputs that emphasize population control, delivery of basic needs and integrated rural development.

Third, Asian governments will have to recognize the reality of these zones of intense urban-rural interaction and direct much of their investment to these areas. This means making hard decisions against fostering small-town development and rural industrialization in less accessible areas without neglecting rural-integrated development schemes in such areas. Many governments should opt for policies of “modified regional growth pole” growth (emphasizing not just the urban pole but also the large mega-urban region of which it is part).

Fourth, Asian governments will need to monitor carefully the growth of economic activities in these zones for the obvious problems that will arise over conflict in incompatible land uses and environmental pollution, for example. This process will involve great care, for one of the major factors favoring the economic vitality of the *desakota* zones is the mixture of land uses.

Fifth, Asian governments will need to improve access in these zones of intense rural-urban interaction with improved roads and fast railway communication. In this respect, the building of the Shinkansen, the Seoul-Pusan Highway, and the Taipei-Kaohsiung Freeway have been crucial to the development of Japan, Korea and Taiwan, respectively. This costly investment reaps developmental rewards.

Finally, Asian governments should develop new spatial systems of data collection similar to those of the “living perimeters” of Taiwan, which will enable them to monitor effectively the impact of investment decisions on labor force composition and income, among others, within the *desakota* zones.

Of course, the timing of government strategies and fiscal ability to implement them are not easy to gauge. However, the demographic and economic reality of the growth of the *desakota* means that planning decisions relating to them cannot be postponed.

Conclusion

The historical evolution of the *desakota* regions has created a unique opportunity for Asian development. Whether this opportunity will be taken advantage of in all Asian countries remains to be seen, but certainly the challenge to urban versus rural growth, which seems so central to many Asian development strategies, can be resolved at least partially through the planned development of *desakota* regions.

Notes

- 1 Jean Gottmann, *Megalopolis: the Urbanized Northeastern Seaboard of the United States* (New York: Kraus, 1961) 257.
- 2 O. H. K. Spate and A. I. A. Learmonth, *India and Pakistan: A General and Regional Geography* (London: Methuen, 1967) 202.
- 3 Some of the ideas in this chapter were first put forward in an address to the International Conference on Asian Urbanization held at the University of Akron in April 1985. See Terry G. McGee, “Urbanisasi or Kotadesasi? Evolving Patterns of Urbanization in Asia,” *Urbanization in Asia: Spatial Dimensions and Policy Issues*, eds. Frank J. Costa, Ashok K. Dutt, Laurence Ma and Allen G. Noble (Honolulu: University of Hawaii Press, 1989) 93-108.
- 4 United Nations, Centre for Human Settlements, *Global Report on Human Settlements* (Oxford: Oxford University Press, 1987).
- 5 Norton Ginsburg, “Extended Metropolitan Regions in Asia: A New Spatial Paradigm,” *The Urban Transition: Reflections on the American and Asian Experiences* (Hong Kong: Chinese University Press, 1990) 21.
- 6 See Terry G. McGee, *Urbanisasi or Kotadesasi? The Emergence of New Regions of Economic Interaction in Asia*, Working Paper 87/8 (Honolulu: EWCEAPI, 1987). See also Norton Ginsburg, *The Urban Transition: Reflections on the American and Asian Experiences* (Hong Kong: Chinese University Press, 1990).
- 7 Jean Gottmann, *Megalopolis: the Urbanized Northeastern Seaboard of the United States* (New York: Twentieth Century Fund, 1961) 257.
- 8 This division between “skill-oriented” and “mechanical” technological inputs is used to buttress Bray’s provocative arguments concerning the distinctive role of Asian wet-rice agriculture in the agrarian development processes in Asia. She argues persuasively that this agroeconomic system has created very different conditions from the “Western model.” This position is certainly central to some of the arguments of this chapter, but its acceptance does not rule out the application of the “mechanical” technological input (as reflected by capital replacing labor in these regions), as can be seen in the growth of agribusiness in the chicken industry in areas such as the Central Plains of Thailand. See Francesca Bray, *The Rice Economies: Technology and Development in Asian Societies* (Oxford: Basil Blackwell, 1986).

- 9 This argument is presented in much greater detail for Latin America in Richard M. Morse, "Trends and Patterns of Latin American Urbanization, 1750-1920," *Comparative Studies in Society and History* 16 (1974) 416-47; and Richard M. Morse, "The Development of Urban Systems in the Americas in the Nineteenth Century," *Journal of Interamerican Studies* 17, 1 (1975) 4-26.
- 10 See Christopher Baker, "Economic Reorganization and the Slump in South and Southeast Asia," *Comparative Studies in Social and Economic History* 23, 2 (1981) 325-49.
- 11 The diagram in Figure 9.1 was constructed by Mike Douglass of the Department of Urban and Regional Planning, University of Hawaii, and myself during the course of some extended evening discussions on the subject of the "urban transition." I am very grateful to him for his constant probing of my ill-formed ideas that formed the basis of earlier presentations of this model. See Mike Douglass, *Urbanization and National Urban Development Strategies in Asia, Indonesia, Korea, and Thailand*, Discussion Paper No. 8 (Honolulu: Department of Urban and Regional Planning, University of Hawaii, 1988).
- 12 The use of a coined Indonesian term taken from the two words *kota* (town) and *desa* (village) was adopted after discussions with Indonesian social scientists, because of my belief that there was a need to look for terms and concepts in the languages of developing countries that reflect the empirical reality of their societies. Reliance solely on the language and concepts of Western social science, which have dominated the analyses of non-Western societies, can lead to a form of "knowledge imperialism." In this text, I have used the term *desakota*, which can be used interchangeably with *kotadesa*.
- 13 The term "peasant" in this context applies not only to those farmers who own their land, but also to tenants operating small units of farmland.
- 14 Robert E. Baldwin, "Patterns of Development in Newly Settled Regions," *The Manchester School of Economic and Social Studies* 24 (1956) 161-79; and Douglas Dowd, "A Comparative Analysis of Economic Development in the American West and South," *Journal of Economic History* 16, 7 (1956) 558-74. See also Richard Morse, "Trends and Patterns"; and Morse, "Urban Systems in the Americas."
- 15 Harry T. Oshima, "The Transition From an Agricultural to an Industrial Economy in East Asia," *Economic Development and Cultural Change* 34, 4 (1986) 783-810.
- 16 Graeme Hugo, *Population Mobility in West Java, Indonesia* (PhD dissertation, Canberra: Australian National University, 1975); Patrick Guinness, *Harmony and Hierarchy in a Javanese Kampong* (Oxford: Oxford University Press, 1986); and Ali Ahmad, *Agricultural Stagnation and Population Pressure: The Case of Bangladesh* (New Delhi: Vikas Publishing House, 1984).
- 17 Yujiro Hayami, *A Century of Agricultural Growth in Japan* (Minneapolis: University of Minnesota Press, 1976); Otohiko Hasumi, "Rural Society in Postwar Japan, Part 1," *The Japan Foundation Newsletter* 12, 5 (1985) 1-10; Otohiko Hasumi, "Rural Society in Postwar Japan, Part 2," *The Japan Foundation Newsletter* 12, 6 (1985) 1-7; Norihiko Nakai, "Urbanization Promotion and Control in Metropolitan Japan," *Planning Perspectives* 3 (1988) 783-810; John Lewis, "Metropolitan Japan," *Planning Perspectives* 3 (1988) 783-810; and John Lewis, "The Real Security Issue: Rice," *Far Eastern Economic Review* (19 June 1981) 70-1.
- 18 Alden Speare, Paul Lie and Ching-lung Tsay, eds. *Urbanization and Development: the Rural-Urban Transition in Taiwan* (Boulder: Westview Press, 1988).
- 19 See Daniel Todd and Yi-Chung Hsueh, "Taiwan: Some Spatial Implications of Rapid Economic Growth," *Geoforum* 19, 2 (1988) 133-45; and Jack F. Williams, "Urban and Regional Planning in Taiwan: the Quest for Balanced Regional Development," *Tijdschrift voor Economische en Social Geografie* 79, 3 (1988) 175-87.
- 20 I am grateful to Rex Casinader and Wang Yaolin, PhD candidates in the Department of Geography at the University of British Columbia, for information on developments in South India and Sichuan.
- 21 An exception to this generalization appears to be the Beijing-Tianjin area, which is a region of intense mixture of crops. Since 1949, the southern part of the Shenyang-Dalian region has become an important area of rice production. The identification of "rice-growing" regions does not preclude the possibility of "mixed crop" systems developing similar population densities. For example, Polly Hill, *Dry Grain Farming Families: Hausaland (Nigeria) and Karnataka (India) Compared* (Cambridge: Cambridge University Press, 1982).
- 22 See Oshima, "The Transition from an Agricultural to an Industrial Economy," 784.
- 23 *Ibid.*, 785.
- 24 Clifford Geertz, *Agricultural Involution: the Process of Ecological Change in Indonesia* (Berkeley: University of California Press, 1963).
- 25 Bray, *Rice Economies*, 135.
- 26 Wang Yaolin's PhD thesis research on the Shenyang-Dalian region is being carried out in the Department of Geography at the University of British Columbia.
- 27 See Mike Douglass, *Regional Integration on the Capitalist Periphery: the Central Plains of Thailand* (The Hague: Institute of Social Studies, 1984); and Mike Douglass, "Population Growth and Policies in Mega-cities: Calcutta," *Population Policy Paper No. 1* (New York: United Nations, 1986).
- 28 Gustav Ranis and Francis Stewart, "Rural Linkages in the Philippines and Thailand," *Macro-policies for Appropriate Technology in Developing Countries*, ed. Frances Stewart (Boulder: Westview Press, 1988) 140-91.
- 29 W. Arthur Lewis, "Economic Development with Unlimited Supplies of Labour," *The Economics of Underdevelopment*, ed. H. N. Agarwala and S. P. Singh (New York: Oxford University Press, 1963) 400-49. See also Geertz, *Agricultural Involution*; and Pierre Gourou, *L'Asie* (Paris: Librairie Hachette, 1953).
- 30 Yue-man Yeung, "Controlling Metropolitan Growth in Eastern Asia," *Geographical Review* 76, 2 (1986) 125-37.
- 31 John Hariss and Mick Moore, "Editors' Introduction to Special Issue on Development and the Rural-Urban Divide," *The Journal of Development Studies* 20, 3 (1984) 1-4.
- 32 Michael Lipton, "Urban Bias Revisited," *The Journal of Development Studies* 20, 3 (1984) 139-66, specifically 155.

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